MIL-T-43113A(AV)
23 April 1981
Superseding
MIL-T-43113(MO)
16 October 1962

MILITARY SPECIFICATION

TOOL SET, STUD REMOVER AND SETTER

This specification is approved for use by US Army Aviation Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a stud and lockring remover and setter tool set.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-40	Packaging and Packing of Hand Tools

MILITARY

MIL-T-704	Treatment and Painting of Materiel
MIL-C-13924	Coating, Oxide, Black, for Ferrous Metals

STANDARDS

FEDERAL

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Research and Development Command, ATTN: DRDAV-ERS, 4300 Goodfellow Boulevard, St. Louis, MO 63120, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5180

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MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Tool set. The tool set shall be a compact unit consisting of metal case, six hollow mills, six wrenches, and six drivers, designed and constructed to provide tools for removing and installing studs and lockrings listed on Figure 1, which specified the sizes required. Weight and dimensions shall be the minimum compatible with intended use. The tools shall be preproduction tested as specified in 4.3.1.

3.2 Materials. The set shall be of a composition and quality of steel which will, when properly heat-treated, produce tools which will function as specified in 3.3.

3.3 Performance.

3.3.1 Hollow mills. When the hollow mills are driven by a standard 1/2-inch portable electric drill, they shall remove the inner serrations of the lockrings and the outer serrations of the studs.

3.3.2 Wrenches. Wrenches shall effectively engage the serrations of the studs used with lockrings listed on Figure 1.

3.3.3 Drivers. Drivers shall fit the outside of the studs, with clearance sufficient to prevent damage to the studs and the surface of the materials in which the lockrings are installed. When the lockrings are installed to the depths reflected in colume C of Figure 1, a machinist's hammer (not to be furnished with this specification) is used to tap the drivers.

3.4 Case. Case material and construction shall enable the tools to be held in their respective compartments without damage if the case is stored in an inverted position. The case shall have metal hinges.

3.5 Identification of product.

3.5.1 Tools. Each tool shall be plainly and permanently marked with the applicable stud size for which it is designed.

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3.5.2 Case. Identification marking shall be in accordance with Standard MIL-STD-130.

3.6 Operator's handbook. One set of operating instructions containing illustrations of the tools and an index shall be provided in each case.

3.7 Finish.

3.7.1 Case. The case shall be treated in accordance with type C of Specification MIL-T-704. Finish coat of paint shall be olive drab conforming to No. 34087 of Standard FED-STD-595.

3.7.2 Hollow mills. Hollow mills shall be oil-treated or surface-treated in a manner to prevent corrosion.

3.7.3 Wrenches and drivers. Wrenches and drivers shall be black oxide conforming to Specification MIL-C-13924.

3.8 Workmanship. Workmanship shall be first class in every respect. Surfaces shall be smooth and free from projections. All tools shall be free from sharp edges, burrs, rust, and any defects which may affect appearance, durability, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. All tool sets, or fraction thereof, offered for delivery at one time, shall be considered a lot for purpose of inspection.

4.2.2 Samples. Sampling shall be in accordance with Standard MIL-STD-105, at inspection level indicated.

4.2.2.1 Case. Case shall be inspected at level II, with acceptable quality level (AQL) of 1.5.

4.2.2.2 Tools. One tool set of each inspection lot shall be inspected as specified in 4.3.1.

4.3 Classification of tests. Tests shall be classified as preproduction and acceptance.

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4.3.1 Preproduction. The first complete tool set manufactured under the production process shall be subjected to all tests specified in 4.5, and shall have the approval of the contracting officer.

4.3.2 Acceptance. Acceptance of the tool set shall be based upon satisfactory compliance of the samples, selected in accordance with 4.2.2, when subjected to the tests specified in 4.4 and 4.5.

4.4 Inspection. Each case and tool selected as specified in 4.2.2.1 and 4.2.2.2 shall be visually inspected for workmanship, finish, marking, and storage of tools and instructions.

4.5 Test procedures.

4.5.1 Hollow mill cutting. Hollow mills selected as specified in 4.2.2.2, and acceptable in accordance with 4.3.2, shall make 10 test cuts of a depth in accordance with the dimensions listed under column C of Figure 1 for the applicable cutter. The material being cut shall be of at least the same hardness as the studs and lockrings. Tests shall be conducted in a stationary drill press of 1/2-inch capacity.

4.5.2 Installation.

4.5.2.1 Wrenches. Wrenches selected as specified in 4.2.2.2, and accepted in accordance with 4.3.2, shall be used for test installation of 10 studs of applicable wrench size.

4.5.2.2 Drivers. Drivers selected in accordance with 4.2.2.2, and in accordance with 4.3.2, shall be subjected to tests equivalent to driving 10 lockrings into position. During this test, material in which the lockrings are installed shall be observed for any damage made by the drivers.

4.6 Preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.

5.1.1 Level A. Tool set shall be cleaned, preserved, and packed in accordance with Federal Specification PPP-P-40.

5.1.2 Level C. Sets shall be prepared in accordance with the supplier's commercial practice in a manner acceptable to common carrier at the lowest acceptable rate.

5.2 Marking. In addition to any special marking required (see 6.3), all individual packages and shipping containers shall be marked for shipment in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. Tool set is intended to remove and set studs with internally and externally serrated lockrings. The lockrings prevent the studs from turning.

6.2 Federal stock number. Federal stock number of the tool set is NSN 5180-778-3789.

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Period of availability of examination and test records (see 4.1).

c. Level of preparation for delivery required (see 5.1.1 and 5.1.2).

d. Whether special marking is required (see 5.2).

6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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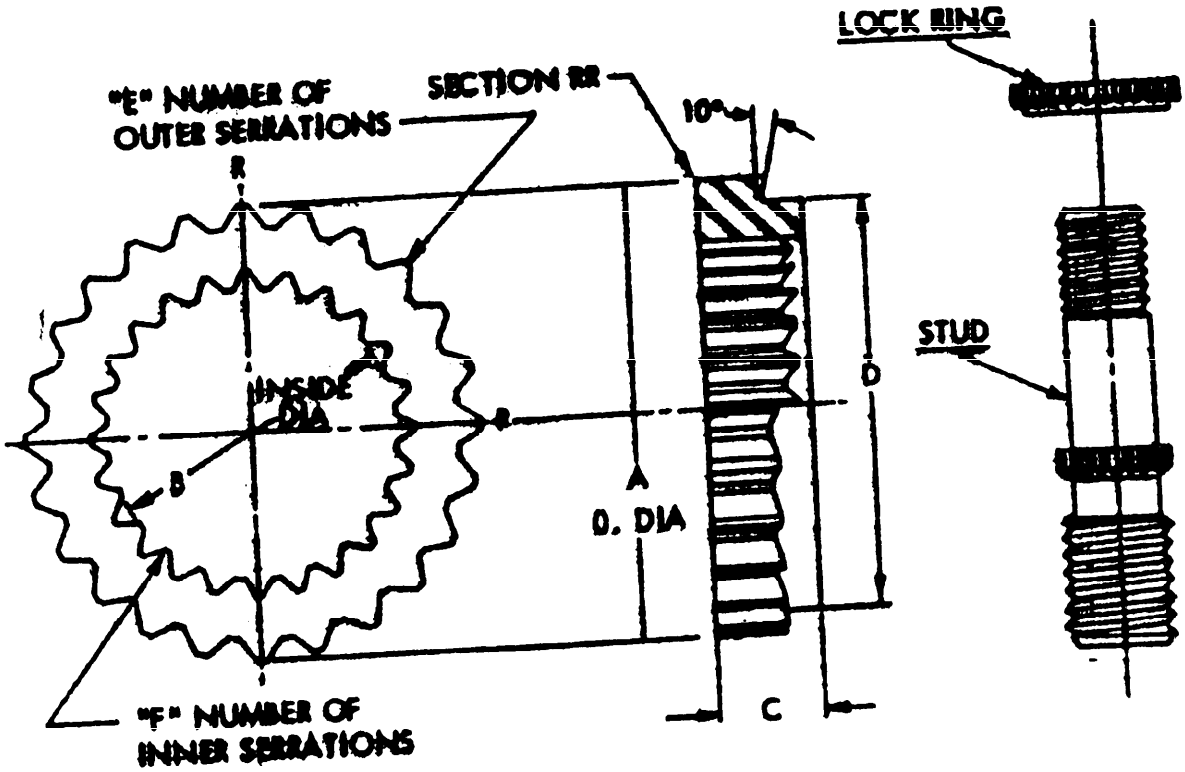
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Lock ring No.	A	B	C	D	E	F	Stud size nut end
1	.354	.213	.099	.298	24	13	10-32
2	.400	.266	.112	.343	24	17	1/4-28
3	.468	.323	.138	.400	26	20	5/16-24
4	.545	.390	.145	.480	34	24	3/8-24
5	.624	.468	.145	.560	36	26	7/16-20
6	.767	.574	.161	.702	40	26	1/2-20

FIGURE 1. -- Lockring Number and Studs Sizes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-43113A TOOL SET, STUD REMOVER & SETTER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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